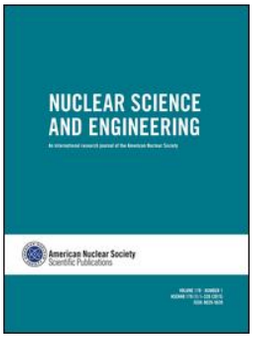

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Laminar Tube Flow with Arbitrary Internal Heat Sources and Wall Heat Transfer

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An analysis is made to determine the heat transfer characteristics for laminar flow of a heat generating fluid in a circular tube with wall heat transfer. The internal heat generation is permitted to vary in an arbitrary manner both longitudinally along the tube and radially across the section. In addition, arbitrary longitudinal variations in the wall heat transfer may be present. The results obtained apply along the entire length of the tube, that is, in the thermal entrance region as well as far down the tube. Numerical results are evaluated for certain special cases such as uniform and parabolic radial heat source distributions.

NOMENCLATURE

A	a constant defined by Eq. (32b)
B	shape parameter in parabolic radial heat source distribution [see Eq. (21)]
C_n	coefficients in series expansion of temperature for uniform radial heat source
c_p	specific heat at constant pressure
D_n	coefficients in series expansion of temperature for parabolic radial heat source
E_n	coefficients in series expansion of temperature for arbitrary radial heat source
F_n	coefficients in series expansion of temperature for uniform wall heat flux
f, f_1, f_2	functions of r
H	function of r defined by Eq. (32a)
k	thermal conductivity
Pr	Prandtl number, $c_p\mu/k = \nu/\alpha$
Q	heat source per unit volume within fluid; Q_0, value of Q at tube centerline
q	local heat flux per unit area at tube wall
R_n	eigenfunctions of Eq. (12)
Re	Reynolds number, $\bar{u}2r_0/\nu$

r	radial coordinate measured from tube centerline; r_0, tube radius
S	arbitrary radial variation in internal heat source
t	temperature; t_0, temperature of fluid entering tube (a constant); t^+ difference temperature defined in Eq. (9)
u	fluid velocity in the x direction; $\bar{u}$, mean fluid velocity in x direction
x	axial coordinate along tube length
x^*	dummy integration variable
α	thermal diffusivity, $k/\rho c_p$
β_n^2	eigenvalues of Eq. (12)
μ	absolute viscosity
ν	kinematic viscosity
ρ	fluid density

SUBSCRIPTS

b	refers to the bulk fluid temperature
fd	designates the fully developed heat transfer condition
Q	refers to situation where $Q \neq 0$ and $q = 0$
q	refers to situation where $q \neq 0$ and $Q = 0$
w	refers to condition at the tube wall

INTRODUCTION

The heat transfer characteristics of systems containing flowing fluids with internal heat generation are of particular interest in nuclear technology, especially in the design of fluid-fueled reactors. Since this type of reactor is frequently constructed with an array of parallel flow passages, one approach to the study of heat generating flows has been to look at the thermal behavior of a typical element such as a single flow passage and its bounding walls. This model has been used by Poppendiek (*1*) and Poppendiek and Palmer (*2, 3, 4*) to compute the *fully-developed* heat transfer in circular tubes, between parallel plates, and in annuli in which the internal heat generation was uniform both across the section and along the length of the passage. Poppendiek (*5*) and Loeffler (*6*) extended the analysis for flow between parallel plates to include variations of the source across the section, while retaining longitudinal uniformity and fully developed conditions. Topper (*7*) obtained results for the thermal entry region in a cylindrical tube with uniform wall temperature and a uniform heat source throughout the tube.

The present investigation is concerned with the more general problem of fluid flow in a circular tube with *arbitrary* variations of internal heat sources in both the radial and longitudinal directions. Furthermore, the heat transfer at the tube wall is permitted to vary longitudinally along the passage in an *arbitrary* manner. The results obtained can be applied along the entire length of the passage, that is, in the thermal entrance region as well as far downstream. The analysis is

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made for *laminar* flow having a fully developed parabolic velocity profile which is unchanging along the tube length.

GENERAL ANALYTICAL CONSIDERATIONS

We begin the problem by writing the equation of energy conservation for fully developed hydrodynamic flow in a circular tube,

$$u \frac{\partial t}{\partial x} = \frac{\alpha}{r} \frac{\partial}{\partial r}\left(r \frac{\partial t}{\partial r}\right) + \frac{Q}{\rho c_p}. \tag{1}$$

The fluid properties have been assumed constant, and viscous dissipation and axial conduction have been neglected compared with radial conduction. The geometrical configuration and coordinate system are shown in Fig. 1, and the symbols are defined in the section on nomenclature. For the general situation considered here, the internal heat source is a prescribed function of both x and r in the form $Q = Q_0(x)S(r)$ and is assumed independent of temperature.[1] The wall heat transfer q is a prescribed function of x. The fluid temperature at the tube entrance is taken to be uniform across the section at a value t_0.

Since the temperature t appears only as the first power in Eq. (1), the simplifications associated with linear equations are at our disposal. In particular, we can separate the general problem into two simpler problems as follows. (1) A situation where there is internal heat generation Q in a tube with insulated walls ($q = 0$). In this case, the temperature distribution is denoted by t_Q. (2) A situation where there is heat transfer q at the walls but no internal heat generation ($Q = 0$). This temperature distribution is designated as t_q. From the linearity of the energy equation we then have for the solution to the general problem

$$t = t_Q + t_q. \tag{2}$$

The equations and boundary conditions used to determine t_Q and t_q are respectively:

$$u \frac{\partial t_Q}{\partial x} = \frac{\alpha}{r} \frac{\partial}{\partial r}\left(r \frac{\partial t_Q}{\partial r}\right) + \frac{Q}{\rho c_p} \tag{3}$$

$$\left.\begin{aligned} \partial t_Q/\partial r &= 0 \quad \text{at } r = r_0 \quad \text{(insulated wall)} \\ \partial t_Q/\partial r &= 0 \quad \text{at } r = 0 \quad \text{(symmetry)} \\ t_Q(0, r) &= t_0 \qquad\qquad \text{(entrance condition)} \end{aligned}\right\} \tag{3a}$$

and

$$u \frac{\partial t_q}{\partial x} = \frac{\alpha}{r} \frac{\partial}{\partial r}\left(r \frac{\partial t_q}{\partial r}\right) \tag{4}$$

[1] Once the solution is known for the heat source distribution $Q = Q_0(x)S(r)$, the linearity of the energy equation permits extension by a simple superposition to the more general form $\sum_i [Q_0(x)S(r)]_i$.

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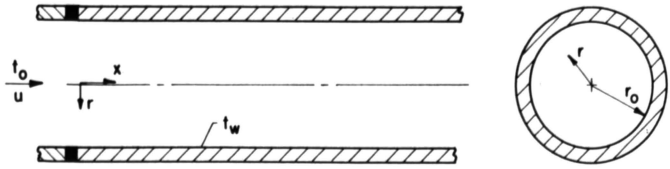

FIG. 1. Physical model and coordinate system.

$$\left.\begin{array}{lll} \partial t_q/\partial r = q/k & \text{at } r = r_0 & \text{(specified heat flux)} \\ \partial t_q/\partial r = 0 & \text{at } r = 0 & \text{(symmetry)} \\ t_q(0, r) = t_0 & & \text{(entrance condition)} \end{array}\right\}. \tag{4a}$$

These two problems will be treated separately and the results combined to yield information for more general situations.

We begin with the problem of a fluid with internal heat sources in an insulated tube. First, the source is taken to be uniform and the solution obtained for both the entrance and fully developed regions. Then radial variations are included and finally longitudinal variations are taken into account.

The problem of wall heat transfer to a fluid having no heat sources is then considered. We first give the results for a uniform wall heat flux and then generalize them to apply for arbitrary longitudinal variations.

INTERNAL HEAT SOURCES WITH TUBE INSULATED

UNIFORM HEAT SOURCE

For this simplest case, the governing equation is obtained by letting Q in Eq. (3) be a constant. The method of solution will be given in some detail, thereby permitting an abbreviated treatment of more complicated cases which are solved later in a similar fashion.

It is well known that as the flow proceeds farther and farther downstream, the heat transfer characteristics approach more and more closely to a limiting condition. This limit is called the fully developed heat transfer situation and will be denoted by the subscript fd. The solution for t_Q is found most easily by separate consideration of the fully developed and entrance regions.

First consider the fully developed temperature $t_{Q,\text{fd}}$. Since conservation of energy must everywhere be satisfied, $t_{Q,\text{fd}}$ obeys Eq. (3),

$$u\,\frac{\partial t_{Q,\text{fd}}}{\partial x} = \frac{\alpha}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial t_{Q,\text{fd}}}{\partial r}\right) + \frac{Q}{\rho c_p}. \tag{5}$$

The fully developed situation for uniform internal heat sources is characterized by the fact that

$$\frac{\partial t_{Q,\text{fd}}}{\partial x} = \text{constant} = \frac{Q}{\rho \bar{u} c_p} \tag{6}$$

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where $\bar{u}$ is the mean flow velocity. Equation (6) states that temperatures at all positions in the tube cross section rise in the same linear fashion along the tube length. For the fully developed situation the boundary condition at the tube entry ($x = 0$) need not be considered (it is accounted for later by the entry region solution) and Eq. (6) can be rephrased as

$$t_{Q,\mathrm{fd}} - t_0 = \frac{Q}{\rho \bar{u} c_p} x + f(r). \tag{6a}$$

The function $f(r)$ is found by inserting Eq. (6a) into the differential equation (5) and using the Poiseuille formula

$$\frac{u}{\bar{u}} = 2\left[1 - \left(\frac{r}{r_0}\right)^2\right] \tag{7}$$

for the velocity distribution. This leads to the equation for $f(r)$

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{df}{dr}\right) = \frac{Q}{k}\left[1 - 2\left(\frac{r}{r_0}\right)^2\right]. \tag{8}$$

This equation, subject to the first two boundary conditions (3a), can be integrated directly. The resulting expression for $t_{Q,\mathrm{fd}}$ is

$$\frac{t_{Q,\mathrm{fd}} - t_0}{(Qr_0^2/k)} = \frac{2(x/r_0)}{\mathrm{RePr}} + \frac{1}{4}\left[\left(\frac{r}{r_0}\right)^2 - \frac{1}{2}\left(\frac{r}{r_0}\right)^4 - \frac{1}{4}\right] \tag{6b}$$

which applies only in the fully developed region downstream of the thermal entrance region.

To determine the solution in the entrance region it is convenient to define a difference temperature t^+ as

$$t_Q^+ = t_Q - t_{Q,\mathrm{fd}}, \quad \text{or} \quad t_Q = t_Q^+ + t_{Q,\mathrm{fd}}. \tag{9(}$$

Since energy conservation still applies, we can introduce the second of Eqs. (9) into Eq. (3). Then, taking account of Eq. (5), it is found that t_Q^+ must satisfy,

$$u\frac{\partial t_Q^+}{\partial x} = \frac{\alpha}{r}\frac{\partial}{\partial r}\left(r\frac{\partial t_Q^+}{\partial r}\right) \tag{10}$$

with boundary conditions $\partial t_Q^+/\partial r = 0$ at $r = 0$ and $r = r_0$. The condition at $x = 0$ will be discussed a little later. It is noted that the heat source does not appear in Eq. (10). We shall try a solution of this equation as a product of two functions, one depending on x alone and the second depending on r alone.

$$\frac{t_Q^+}{(Qr_0^2/k)} = \chi(x)R(r). \tag{11}$$

Substituting into Eq. (10), and solving for $\chi(x)$ we find that t_Q^+ can be expressed

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by the series

$$\frac{t_Q^+}{(Qr_0^2/k)} = \sum_{n=1}^{\infty} C_n R_n \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\left(\frac{x}{r_0}\right)\right] \tag{11a}$$

β_n^2 and R_n are respectively the eigenvalues and eigenfunctions of the following Sturm-Liouville problem:

$$\begin{gathered}\frac{d^2 R_n}{d(r/r_0)^2} + \frac{r_0}{r}\frac{dR_n}{d(r/r_0)} + R_n \beta_n^2\left[1 - \left(\frac{r}{r_0}\right)^2\right] = 0 \\ \frac{dR_n}{d(r/r_0)} = 0 \qquad \text{at } r = 0 \qquad \text{and } r = r_0 .\end{gathered} \tag{12}$$

The first seven eigenvalues and eigenfunctions of Eq. (12) have been computed (using an IBM 653 computer). Values of β_n^2 and R_n (1) are given in Table I.

The coefficients C_n in Eq. (11a) still remain to be determined. These are evaluated to satisfy the third boundary condition of Eq. (3a) which specifies that at the entrance of the tube

$$t_Q(0, r) = t_0 = t_Q^+(0, r) + t_{Q,\mathrm{fd}}(0, r). \tag{13}$$

Evaluating Eqs. (6b) and (11a) at $x = 0$ and substituting into Eq. (13) gives

$$\frac{t_Q^+(0, r)}{(Qr_0^2/k)} = \sum_{n=1}^{\infty} C_n R_n = -\frac{1}{4}\left[\left(\frac{r}{r_0}\right)^2 - \frac{1}{2}\left(\frac{r}{r_0}\right)^4 - \frac{1}{4}\right]. \tag{14}$$

Using the properties of the Sturm-Liouville system, we have immediately that

$$C_n = -\frac{\frac{1}{4}\int_0^1 \left[\left(\frac{r}{r_0}\right)^2 - \frac{1}{2}\left(\frac{r}{r_0}\right)^4\right]\left[\left(\frac{r}{r_0}\right) - \left(\frac{r}{r_0}\right)^3\right] R_n \, d\left(\frac{r}{r_0}\right)}{\int_0^1 \left[\left(\frac{r}{r_0}\right) - \left(\frac{r}{r_0}\right)^3\right] R_n^2 \, d\left(\frac{r}{r_0}\right)} . \tag{15}$$

Values of C_n are given in Table I.

The complete solution for the temperature which applies in both the entrance

TABLE I

n	β_n^2	$R_n(1)$	C_n	F_n
1	25.6796	−0.492517	0.0799573	0.403483
2	83.8618	0.395508	−0.0254050	−0.175111
3	174.167	−0.345872	0.0124496	0.105594
4	296.536	0.314047	−0.00738848	−0.0732804
5	450.947	−0.291252	0.00489305	0.0550357
6	637.387	0.273808	−0.0034795	−0.043483
7	855.850	−0.259852	0.0026019	0.035597

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and fully developed regions is found by adding the solutions for $t_{Q,\mathrm{fd}}$ and t_Q^+, giving

$$\frac{t_Q - t_0}{(Qr_0^2/k)} = \frac{2(x/r_0)}{\mathrm{RePr}} + \frac{1}{4}\left[\left(\frac{r}{r_0}\right)^2 - \frac{1}{2}\left(\frac{r}{r_0}\right)^4 - \frac{1}{4}\right] + \sum_{n=1}^{\infty} C_n R_n \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\left(\frac{x}{r_0}\right)\right]. \tag{16}$$

The result which is usually of practical interest is the wall temperature variation corresponding to the uniform heat source. Setting $r = r_0$ in Eq. (16) and using the subscript w to denote wall conditions, we obtain

$$\frac{t_{Q,w} - t_0}{(Qr_0^2/k)} = \frac{2(x/r_0)}{\mathrm{RePr}} + \frac{1}{16} + \sum_{n=1}^{\infty} C_n R_n(1) \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\left(\frac{x}{r_0}\right)\right]. \tag{17}$$

Another form of this equation which may be more convenient in some instances is obtained by introducing the bulk temperature, $t_{Q,b}$. For a uniform heat source, the bulk temperature is given by

$$t_{Q,b} = t_0 + \frac{2Qr_0^2}{k}\frac{(x/r_0)}{\mathrm{RePr}}. \tag{18}$$

Then, for the fully developed situation

$$(t_{Q,w} - t_{Q,b})_{\mathrm{fd}} = \frac{1}{16}\frac{Qr_0^2}{k}. \tag{19}$$

Combining Eqs. (17) and (18) and then dividing by Eq. (19) yields the important ratio,

$$\frac{t_{Q,w} - t_{Q,b}}{(t_{Q,w} - t_{Q,b})_{\mathrm{fd}}} = 1 + 16\sum_{n=1}^{\infty} C_n R_n(1) \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\left(\frac{x}{r_0}\right)\right]. \tag{20}$$

This relation has been evaluated using the numerical data of Table I and a plot is given on Fig. 2.

We can define a thermal entrance length as the heated length required for $(t_{Q,w} - t_{Q,b})$ to approach within 5% of the fully developed value. From Fig. 2 the value of $(x/r_0)/\mathrm{RePr}$ corresponding to an ordinate of 0.95 is approximately 0.10. Then, for example, if RePr is 1000, the five per cent entrance length is 100 r_0 (50 diameters).

Parabolic Radial Source Variations, Longitudinally Uniform

Next we consider the case where the internal heat sources vary in a parabolic manner along the tube radius, while longitudinal uniformity is preserved. Then the source distribution has the form

$$Q = Q_0\left[1 + B\left(\frac{r}{r_0}\right)^2\right] \tag{21}$$

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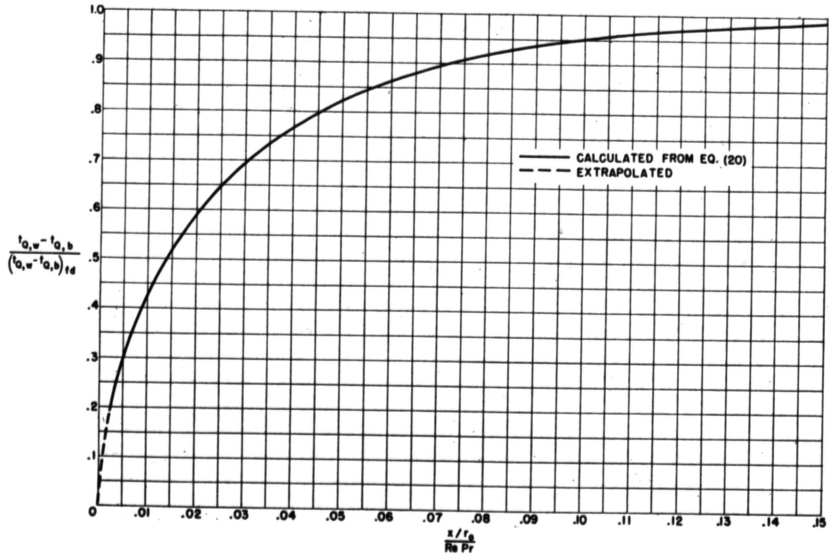

FIG. 2. Dimensionless wall temperature variation for uniform internal heat source in insulated tube.

where Q_0 (the centerline heat source) and B are constants which may be chosen arbitrarily. For $B = 0$, Eq. (21) reduces to the uniform heat source situation which has been treated in the previous section.

The solution proceeds in the same manner as for the uniform source case, that is, we first find the fully developed solution and then deal with a difference temperature. To determine the fully developed temperature distribution equation (5) still applies, but now we replace Q with the distribution given in Eq. (21),

$$u \frac{\partial t_{Q,\mathrm{fd}}}{\partial x} = \frac{\alpha}{r} \frac{\partial}{\partial r}\left(r \frac{\partial t_{Q,\mathrm{fd}}}{\partial r}\right) + \frac{Q_0}{\rho c_p}\left[1 + B\left(\frac{r}{r_0}\right)^2\right]. \tag{22}$$

The condition for fully developed heat transfer which now replaces Eqs. (6) and (6a) is

$$\frac{\partial t_{Q,\mathrm{fd}}}{\partial x} = \frac{Q_0(1 + B/2)}{\rho \bar{u} c_p} \tag{23}$$

or

$$t_{Q,\mathrm{fd}} - t_0 = \frac{Q_0(1 + B/2)x}{\rho \bar{u} c_p} + f_1(r). \tag{23a}$$

If we substitute this relation into Eq. (22) and use the Poiseuille distribution for u as given by Eq. (7), we obtain a differential equation for the function $f_1(r)$.

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This equation can be immediately integrated subject to the first two boundary conditions in Eq. (3a). The function $f_1(r)$ is then incorporated into Eq. (23a) to yield the solution for $t_{Q,\mathrm{fd}}$ as follows:

$$\frac{t_{Q,\mathrm{fd}} - t_0}{\left(\frac{Q_0 r_0^2}{k}\right)\left(1 + \frac{B}{2}\right)} = \frac{2(x/r_0)}{\mathrm{RePr}} + \frac{2B + 2}{B + 2}\left\{\frac{1}{4}\left[\left(\frac{r}{r_0}\right)^2 - \frac{1}{2}\left(\frac{r}{r_0}\right)^4 - \frac{1}{4}\right]\right\}. \quad (24)$$

When $B = 0$ this expression reduces to Eq. (6b) for uniform radial heat sources. A noteworthy feature of Eq. (24) is that the shape of the radial temperature profile is independent of B!

To determine the temperature distribution in the entrance region we proceed as before by defining a difference temperature t_Q^+ as in Eq. (9). The difference temperature again satisfies Eq. (10) and the solution can be written as

$$\frac{t_Q^+}{\left(\frac{Q_0 r_0^2}{k}\right)\left(1 + \frac{B}{2}\right)} = \sum_{n=1}^{\infty} D_n R_n \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\left(\frac{x}{r_0}\right)\right] \quad (25)$$

where β_n and R_n are the same eigenvalues and eigenfunctions obtained previously from Eq. (12). The coefficients D_n are found by applying the boundary condition at $x = 0$ as given in Eq. (13). Evaluating $t_{Q,\mathrm{fd}}(0, r)$ from Eq. (24) and $t_Q^+(0, r)$ from Eq. (25), this boundary condition becomes

$$\frac{t_Q^+(0, r)}{\left(\frac{Q_0 r_0^2}{k}\right)\left(1 + \frac{B}{2}\right)} = \sum_{n=1}^{\infty} D_n R_n = -\frac{2B + 2}{B + 2}\left\{\frac{1}{4}\left[\left(\frac{r}{r_0}\right)^2 - \frac{1}{2}\left(\frac{r}{r_0}\right)^4 - \frac{1}{4}\right]\right\}. \quad (26)$$

Comparing Eqs. (26) and (14), it is evident that D_n and C_n are related by

$$D_n = \frac{2B + 2}{B + 2} C_n. \quad (27)$$

Due to this fortunate relation, we are not required to carry out further numerical computations to determine the D_n.

The complete solution for the temperature which applies at any position along the tube is obtained by adding the relations for $t_{Q,\mathrm{fd}}$ and t_Q^+ giving the result,

$$\begin{aligned}\frac{t_Q - t_0}{\left(\frac{Q_0 r_0^2}{k}\right)\left(1 + \frac{B}{2}\right)} &= \frac{2(x/r_0)}{\mathrm{RePr}} \\ &+ \frac{2B + 2}{B + 2}\left\{\frac{1}{4}\left[\left(\frac{r}{r_0}\right)^2 - \frac{1}{2}\left(\frac{r}{r_0}\right)^4 - \frac{1}{4}\right] + \sum_{n=1}^{\infty} C_n R_n \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\left(\frac{x}{r_0}\right)\right]\right\}.\end{aligned} \quad (28)$$

To find the wall temperature $t_{Q,w}$ it is only necessary to set $r = r_0$. Then, the

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equation can be rephrased in terms of the bulk temperature $t_{Q,b}$ with the result

$$\frac{t_{Q,w} - t_{Q,b}}{(t_{Q,w} - t_{Q,b})_{\mathrm{fd}}} = 1 + 16 \sum_{n=1}^{\infty} C_n R_n(1) \exp\left[- \frac{\beta_n^2}{\mathrm{RePr}} \left(\frac{x}{r_0}\right)\right] \tag{29}$$

where

$$(t_{Q,w} - t_{Q,b})_{\mathrm{fd}} = \frac{B + 1}{16} \frac{Q_0 r_0^2}{k} \tag{29a}$$

and

$$t_{Q,b} = t_0 + \frac{B + 2}{\mathrm{RePr}} \frac{Q_0 r_0^2}{k} \frac{x}{r_0}. \tag{29b}$$

It is noted that Eq. (29) is identical to Eq. (20). So, the plot of the temperature ratio as given in Fig. 2 applies to both radially uniform and to parabolic radial source variations. The distinction between the two cases arises when different values of B are introduced into Eqs. (29a) and (29b).

Arbitrary Radial Source Variations, Longitudinally Uniform

In this section we consider the internal source to be of the form

$$Q = Q_0 S(r) \tag{30}$$

where $S(r)$ is any arbitrary function of r, and Q_0 is a constant which is selected as the value of Q at $r = 0$. We proceed as before, working first with the fully developed situation and then with a difference temperature in the entrance region.

The fully developed situation is now characterized by

$$\frac{\partial t_{Q,\mathrm{fd}}}{\partial x} = \frac{2Q_0 \int_0^{r_0} rS\, dr}{\rho \bar{u} c_p r_0^2} \tag{31}$$

or

$$t_{Q,\mathrm{fd}} - t_0 = \left(\frac{2Q_0 \int_0^{r_0} rS\, dr}{\rho \bar{u} c_p r_0^2} \right) x + f_2(r). \tag{31a}$$

The function $f_2(r)$ is found, as in the previous cases, by applying the equation of energy conservation, Eq. (5). This gives the following fully developed temperature profile,

$$\frac{t_{Q,\mathrm{fd}} - t_0}{\dfrac{2Q_0}{k} \displaystyle\int_0^{r_0} rS\, dr} = \frac{2(x/r_0)}{\mathrm{RePr}} + \frac{1}{2}\left[\left(\frac{r}{r_0}\right)^2 - \frac{1}{4}\left(\frac{r}{r_0}\right)^4\right] + H(r) + A \tag{32}$$

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where

$$H(r) = -\frac{\int_0^r \left(\frac{1}{r}\int_0^r rS\,dr\right)dr}{2\int_0^{r_0} rS\,dr} \tag{32a}$$

and

$$A = -4\left(\frac{1}{r_0^2}\int_0^{r_0} rH\,dr - \frac{1}{r_0^4}\int_0^{r_0} r^3H\,dr + \frac{7}{192}\right). \tag{32b}$$

For the entrance region we again use the difference temperature t_Q^+ of Eq. (9). Following the same pattern of steps as in the previous cases we find the solution for t_Q^+ to be

$$\frac{t_Q^+}{\frac{2Q_0}{k}\int_0^{r_0} rS\,dr} = \sum_{n=1}^{\infty} E_n R_n \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\left(\frac{x}{r_0}\right)\right] \tag{33}$$

where again the β_n and R_n are the eigenvalues and eigenfunctions of Eq. (12). The coefficients E_n are found by applying the boundary condition at $x = 0$ and using the properties of the Sturm-Liouville system with the result

$$E_n = -\frac{\int_0^1 \left[H + \frac{1}{2}\left(\frac{r}{r_0}\right)^2 - \frac{1}{8}\left(\frac{r}{r_0}\right)^4\right]\left[\frac{r}{r_0} - \left(\frac{r}{r_0}\right)^3\right] R_n\, d\left(\frac{r}{r_0}\right)}{\int_0^1 \left[\frac{r}{r_0} - \left(\frac{r}{r_0}\right)^3\right] R_n^2\, d\left(\frac{r}{r_0}\right)}. \tag{34}$$

When the radial variation $S(r)$ is specified, $H(r)$ can be computed from Eq. (32a) and the E_n found from Eq. (34). This latter computation requires a knowledge of the eigenfunctions R_n. Space limitations preclude an extensive listing of the eigenfunctions in this paper, but they are available on IBM cards.

The final expression for the temperature distribution at any location in the tube corresponding to an arbitrary radial source distribution is obtained by summing $t_{Q,\mathrm{fd}}$ and t_Q^+, giving

$$\begin{aligned}\frac{t_Q - t_0}{\frac{2Q_0}{k}\int_0^{r_0} rS\,dr} &= \frac{2(x/r_0)}{\mathrm{RePr}} + \frac{1}{2}\left[\left(\frac{r}{r_0}\right)^2 - \frac{1}{4}\left(\frac{r}{r_0}\right)^4\right] \\ &\quad + H(r) + A + \sum_{n=1}^{\infty} E_n R_n \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\left(\frac{x}{r_0}\right)\right].\end{aligned} \tag{35}$$

The wall temperature along the length of the tube can be conveniently expressed in the following ratio, obtained by setting $r = r_0$ and introducing the

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fluid bulk temperature,

$$\frac{t_{Q,w} - t_{Q,b}}{(t_{Q,w} - t_{Q,b})_{\mathrm{fd}}} = 1 + \frac{\sum_{n=1}^{\infty} E_n R_n(1) \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\left(\frac{x}{r_0}\right)\right]}{\frac{3}{8} + H(r_0) + A} \tag{36}$$

where

$$(t_{Q,w} - t_{Q,b})_{\mathrm{fd}} = \left(\frac{2Q_0}{k}\int_0^{r_0} rS\,dr\right)\left[\frac{3}{8} + H(r_0) + A\right] \tag{36a}$$

and

$$t_{Q,b} = t_0 + \frac{4Q_0}{k\,\mathrm{RePr}}\frac{x}{r_0}\int_0^{r_0} rS\,dr. \tag{36b}$$

Arbitrary Longitudinal Heat Source Variations

We will now generalize the results of the three preceding sections to apply for arbitrary longitudinal variations of internal heat source.

First, consider the case where the heat source is uniform radially across the tube, but has an arbitrary axial variation $Q(x)$ as shown schematically in Fig. 3. Envision for the moment an elementary heating process in which there is no heat generation up to a position x^*, and at that position there begins an internal uniform heat generation of an amount dQ. The wall temperature response for such a process is found from Eq. (17) to be

$$\frac{t_{Q,w} - t_0}{r_0^2/k} = dQ\left(\frac{2\frac{x - x^*}{r_0}}{\mathrm{RePr}} + \frac{1}{16} + \sum_{n=1}^{\infty} C_n R_n(1) \exp\left[-\frac{\beta_n^2}{\mathrm{RePr}}\frac{x - x^*}{r_0}\right]\right), \tag{37}$$

$$x \geqq x^*.$$

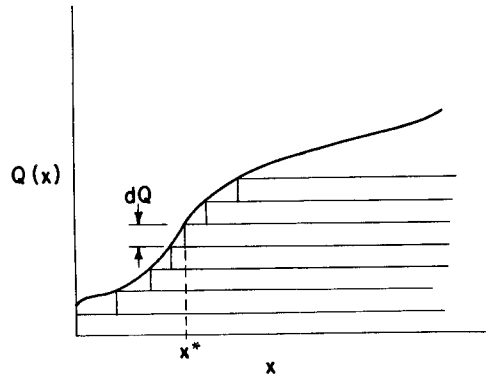

Fig. 3. Representation of arbitrary longitudinal internal heat source distribution.

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This elementary heating may be considered as a differential step in the heat source variation pictured on Fig. 3. By adding up (integrating) the effect of these elementary steps it is possible to determine the wall temperature at any position for a prescribed upstream variation in internal heat generation. The integral of the right side of Eq. (37) can be put in a more convenient form by integrating by parts. This yields the final result,

$$t_{Q,w} - t_{Q,b} = -\frac{r_0/k}{\mathrm{RePr}} \sum_{n=1}^{\infty} C_n \beta_n^2 R_n(1) \int_0^x Q(x^*) \exp\left(-\frac{\beta_n^2}{\mathrm{RePr}} \frac{x - x^*}{r_0}\right) dx^* \quad (38)$$

where

$$t_{Q,b} - t_0 = \frac{2(r_0/k)}{\mathrm{RePr}} \int_0^x Q(x^*)\, dx^*. \quad (38a)$$

Values of C_n, β_n^2, and $R_n(1)$, needed to obtain numerical results, are given in Table I.

Next, consider the case of a parabolic radial source distribution with an arbitrary longitudinal variation written as follows

$$Q = Q_0(x)\left[1 + B\left(\frac{r}{r_0}\right)^2\right]. \quad (39)$$

Using the reasoning of the preceding paragraph, the result given by Eq. (28) for constant Q_0 can be generalized to apply when Q_0 is an arbitrary function of x. The resulting expression for the wall temperature is

$$t_{Q,w} - t_{Q,b} = -\frac{r_0/k}{\mathrm{RePr}} (B + 1) \sum_{n=1}^{\infty} C_n \beta_n^2 R_n(1) \cdot \int_0^x Q(x^*) \exp\left(-\frac{\beta_n^2}{\mathrm{RePr}} \frac{x - x^*}{r_0}\right) dx^* \quad (40)$$

where

$$t_{Q,b} - t_0 = \frac{(r_0/k)(B + 2)}{\mathrm{RePr}} \int_0^x Q(x^*)\, dx^*. \quad (40a)$$

Finally, we consider the more general situation of arbitrary source distributions in both the radial and longitudinal directions which can be written as

$$Q = Q_0(x)S(r). \quad (41)$$

The wall temperature result for arbitrary $S(r)$ and constant Q_0 has already been given in Eq. (36). The generalization for arbitrary $Q_0(x)$ is accomplished by the superposition procedure used in the preceding paragraphs, giving the result

$$t_{Q,w} - t_{Q,b} = -\frac{2}{r_0 k\, \mathrm{RePr}} \left[\int_0^{r_0} rS\, dr\right] \cdot \left[\sum_{n=1}^{\infty} E_n \beta_n^2 R_n(1) \int_0^x Q(x^*) \exp\left(-\frac{\beta_n^2}{\mathrm{RePr}} \frac{x - x^*}{r_0}\right) dx^*\right] \quad (42)$$

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where

$$t_{Q,b} - t_0 = \frac{4}{kr_0 \operatorname{RePr}} \left[\int_0^{r_0} rS\, dr \right] \left[\int_0^x Q(x^*)\, dx^* \right]. \tag{42a}$$

Equations (42) and (42a) give the complete solution for the wall temperature variation corresponding to arbitrarily prescribed radial and longitudinal internal heat source variations in an insulated tube.

WALL HEAT TRANSFER WITHOUT INTERNAL SOURCES

The wall temperature in a circular tube with prescribed wall flux q and without internal heat sources is given in another paper (*8*). For use in a later section, only the final results will be given here.

Uniform Heat Flux

When the wall heat flux q is uniform along the tube length, the variation in wall temperature $t_{q,w}$ is given (*8*) as

$$\frac{t_{q,w} - t_0}{q \dfrac{r_0}{k}} = \frac{4(x/r_0)}{\operatorname{RePr}} + \frac{11}{24} + \sum_{n=1}^{\infty} F_n R_n(1) \exp\left[-\frac{\beta_n^2}{\operatorname{RePr}} \left(\frac{x}{r_0} \right) \right] \tag{43}$$

where the β_n^2 and $R_n(1)$ are the same eigenvalues and eigenfunctions which have already been listed in Table I in connection with the internal heat source problem. The coefficients F_n, needed to evaluate Eq. (43), are listed in the last column of Table I.

Equation (43) can be rephrased in a somewhat more convenient form by introducing the bulk temperature $t_{q,b}$. Then,

$$\frac{t_{q,w} - t_{q,b}}{(t_{q,w} - t_{q,b})_{\mathrm{fd}}} = 1 + \frac{24}{11} \sum_{n=1}^{\infty} F_n R_n(1) \exp\left[-\frac{\beta_n^2}{\operatorname{RePr}} \left(\frac{x}{r_0} \right) \right] \tag{44}$$

where

$$(t_{q,w} - t_{q,b})_{\mathrm{fd}} = \frac{11}{24} \frac{qr_0}{k} \tag{44a}$$

and

$$t_{q,b} = t_0 + \frac{4}{\operatorname{RePr}} \frac{qr_0}{k} \frac{x}{r_0}. \tag{44b}$$

Using the numerical data of Table I, Eq. (44) has been plotted on Fig. 4 This curve gives the wall temperature for the situation of uniform wall heat flux and no internal heat sources. We can define the thermal entrance length as the heated length in which $(t_{q,w} - t_{q,b})$ approaches to within 5% of the fully developed value. From Fig. 4, it is found that the thermal entrance length is given by $(x/r_0)/\operatorname{RePr} = 0.085$.

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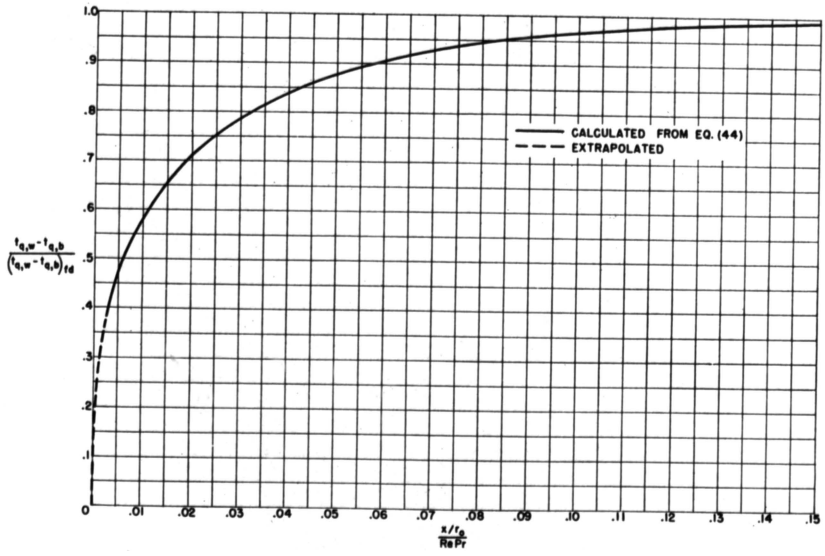

FIG. 4. Dimensionless wall temperature variation for uniform heat flux at tube wall; no internal heat source.

ARBITRARY WALL HEAT FLUX

The results for uniform wall heat flux have been generalized to the case where the heat flux varies arbitrarily along the tube length (*8*). The result for arbitrary $q(x)$ is given as

$$t_{q,w} - t_{q,b} = -\frac{4}{k\,\mathrm{RePr}} \sum_{n=1}^{\infty} F_n \beta_n^2 R_n(1) \int_0^x q(x^*) \exp\left(-\frac{\beta_n^2}{\mathrm{RePr}} \frac{x - x^*}{r_0}\right) dx^* \quad (45)$$

where

$$t_{q,b} - t_0 = -\frac{4}{k\,\mathrm{RePr}} \int_0^x q(x^*)\, dx^* \quad (45a)$$

Values of β_n^2, $R_n(1)$, and F_n are given in Table I.

COMBINED INTERNAL SOURCES AND WALL HEAT TRANSFER

Results for the situation where internal heat generation and wall heat transfer are occurring simultaneously can be found, according to Eq. (2), by adding the contributions due to each. The only caution to be exercised is to make certain that the superposition does not give an incorrect value for the temperature of the entering fluid.[2]

[2] For instance, if the fluid for a combined heat source and wall transfer problem enters at t_0, then it is evident from Eq. (2) that the separate problems for t_Q and t_q cannot both have entering temperatures of t_0.

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The most important result of practical interest is the variation in wall temperature corresponding to a given heating condition. We proceed to determine this by applying Eq. (2) to the wall and bulk temperatures,

$$t_w = t_{Q,w} + t_{q,w} \qquad t_b = t_{Q,b} + t_{q,b} \,. \tag{46}$$

Subtracting the second of these from the first, there is obtained

$$t_w - t_b = (t_{Q,w} - t_{Q,b}) + (t_{q,w} - t_{q,b}). \tag{47a}$$

Let us first discuss the simpler situation where there is no axial variation in Q or q. Then a more convenient form of Eq. (47a) is

$$t_w - t_b = \left[\frac{t_{Q,w} - t_{Q,b}}{(t_{Q,w} - t_{Q,b})_{\mathrm{fd}}}\right](t_{Q,w} - t_{Q,b})_{\mathrm{fd}} + \left[\frac{t_{q,w} - t_{q,b}}{(t_{q,w} - t_{q,b})_{\mathrm{fd}}}\right](t_{q,w} - t_{q,b})_{\mathrm{fd}} \tag{47b}$$

where as before fd means fully developed. If the internal heat sources are uniformly or parabolically distributed over the radius, the numerical values of the first square bracket are given in Fig. 2 or Eqs. (20) or (29). For arbitrary radial variations in internal heat source the first square bracket can be computed from Eq. (36). Numerical values of the ratio in the second square bracket are immediately available from Fig. 4 or Eq. (44). The fully developed temperature differences, $(t_{Q,w} - t_{Q,b})_{\mathrm{fd}}$ and $(t_{q,w} - t_{q,b})_{\mathrm{fd}}$ are given in Eqs. (19), (29a), (36a), and (44a).

For combined internal sources and wall heat transfer which vary in the longitudinal as well as the radial direction, we use Eq. (47a). The expression for $(t_{q,w} - t_{q,b})$ may be evaluated from Eq. (45) for any prescribed variation of q with x. For $(t_{Q,w} - t_{Q,b})$, we use Eqs. (38), (40), or (42) depending on whether the source is radially uniform, parabolic, or arbitrary.

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